

1. Create and manage user groups.

Objective: To create, modify, and manage user groups in Linux and assign users to groups.

Purpose: Linux groups are used to manage permissions for multiple users. Instead of assigning permissions individually to every user, users can be added to a group and permissions can be applied to that group.

Implementation:

```
sunith@sunith-virtual-machine:~$ sudo groupadd techie
[sudo] password for sunith:
sunith@sunith-virtual-machine:~$ sudo groupdel techie
sunith@sunith-virtual-machine:~$ useradd alice
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
sunith@sunith-virtual-machine:~$ sudo useradd alice
sunith@sunith-virtual-machine:~$ sudo groupadd techie
sunith@sunith-virtual-machine:~$ sudo usermod -aG techie alice
sunith@sunith-virtual-machine:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),1002(techie)
sunith@sunith-virtual-machine:~$ sudo groupmod -n dev-team techie
sunith@sunith-virtual-machine:~$ sudo groupmod -d alice dev-team
groupmod: invalid option -- 'd'
Usage: groupmod [options] GROUP

Options:
  -g, --gid GID             change the group ID to GID
  -h, --help                display this help message and exit
  -n, --new-name NEW_GROUP  change the name to NEW_GROUP
  -o, --non-unique          allow to use a duplicate (non-unique) GID
  -p, --password PASSWORD  change the password to this (encrypted)
                           PASSWORD
  -R, --root CHROOT_DIR    directory to chroot into
  -P, --prefix PREFIX_DIR  prefix directory where are located the /etc/* files

sunith@sunith-virtual-machine:~$ ^C
sunith@sunith-virtual-machine:~$ sudo gpasswd -d alice dev-team
Removing user alice from group dev-team
sunith@sunith-virtual-machine:~$ sudo groupdel techie
groupdel: group 'techie' does not exist
sunith@sunith-virtual-machine:~$ sudo groupdel dev-team
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully Implementation is Completed

2. Set up SSH password less authentication.

Objective: To configure SSH passwordless authentication between a client Linux machine and a remote Linux server using SSH keys.

Purpose: Passwordless SSH allows users or administrators to securely connect to remote servers without entering a password every time. It is commonly used for server administration, automation, scripting, and DevOps tools.

Implementation:

```
HP@Ashish MINGW64 ~
$ ssh -i ~/.ssh/id_rsa sunith@192.168.181.128
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

4 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Mon Aug 17 23:06:52 2026 from 192.168.181.1
sunith@sunith-virtual-machine:~$ pwd
/home/sunith
sunith@sunith-virtual-machine:~$ whoami
sunith
sunith@sunith-virtual-machine:~$ |
```


Implementation:

```
sunith@sunith-virtual-machine:~$  
sunith@sunith-virtual-machine:~$ echo h i j k l m >> techie  
sunith@sunith-virtual-machine:~$
```

```
sunith@sunith-virtual-machine:~$ vim techie  
sunith@sunith-virtual-machine:~$ tail -f techie  
a  
b  
c  
d  
e  
f  
g  
h i j k l m
```

Conclusion: Sucessfully Implementation is Completed

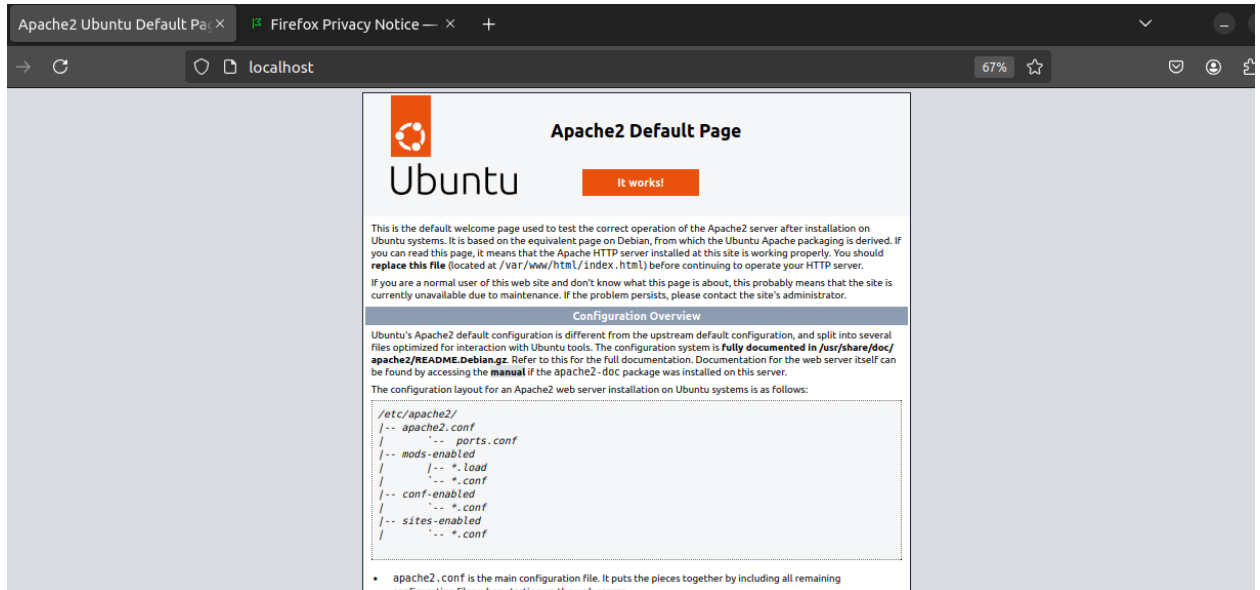
4. Set up a web server (e.g., Apache or Nginx).

Objective: To install and configure a web server such as Apache or Nginx on a Linux system and serve a web page.

Purpose: A web server receives HTTP/HTTPS requests from clients and provides web content such as HTML pages, images, CSS, JavaScript, and application responses.

Implementation:

```
sunith@sunith-virtual-machine:~$ sudo apt install apache2  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
apache2 is already the newest version (2.4.52-1ubuntu4.23).  
The following packages were automatically installed and are no longer required:  
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsmbios-c2  
Use 'sudo apt autoremove' to remove them.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
sunith@sunith-virtual-machine:~$ sudo systemctl start apache2  
sunith@sunith-virtual-machine:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)  
   Active: active (running) since Mon 2026-08-17 22:52:21 IST; 55min ago  
     Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 893 (apache2)  
     Tasks: 55 (limit: 3389)  
    Memory: 7.8M  
       CPU: 396ms  
    CGroup: /system.slice/apache2.service  
            └─893 /usr/sbin/apache2 -k start  
              └─902 /usr/sbin/apache2 -k start  
                └─903 /usr/sbin/apache2 -k start  
  
Aug 17 22:52:20 sunith-virtual-machine systemd[1]: Starting The Apache HTTP Server...  
Aug 17 22:52:21 sunith-virtual-machine apachectl[874]: AH00558: apache2: Could not reliably determine the server's  
Aug 17 22:52:21 sunith-virtual-machine systemd[1]: Started The Apache HTTP Server.  
lines 1-16/16 (END)
```



Conclusion: Successfully Implementation is completed

5. Configure and secure a MySQL Database.

Objective: To install, configure, and secure a MySQL/MariaDB database server on Linux.

Purpose: MySQL/MariaDB is used to store and manage structured application data. Database security is important to prevent unauthorized access and protect sensitive information.

Implementation:

```
sunith@sunith-virtual-machine:~$ sudo apt install mariadb-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  galera-4 gawk libcgi-fast-perl libcgi-pm-perl libconfig-inifiles-perl libdbd-mysql-p
  libmysqlclient21 libsigsegv2 libsnappy1v5 libterm-readkey-perl liburing2 mariadb-cli
  mysql-common socat
Suggested packages:
  gawk-doc libmldbm-perl libnet-daemon-perl libsql-statement-perl libipc-sharedcache-p
The following NEW packages will be installed:
  galera-4 gawk libcgi-fast-perl libcgi-pm-perl libconfig-inifiles-perl libdbd-mysql-p
  libmysqlclient21 libsigsegv2 libsnappy1v5 libterm-readkey-perl liburing2 mariadb-cli
```

```
sunith@sunith-virtual-machine:~$ sudo mysql_secure_installation
```

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
SERVERS IN PRODUCTION USE! PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):

OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n

```
sunith@sunith-virtual-machine:~$ sudo systemctl start mariab
Failed to start mariab.service: Unit mariab.service not found.
sunith@sunith-virtual-machine:~$ sudo systemctl start mariadb
sunith@sunith-virtual-machine:~$ sudo systemctl enable mariadb
Synchronizing state of mariadb.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mariadb
sunith@sunith-virtual-machine:~$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.6.23 database server
   Loaded: loaded (/lib/systemd/system/mariadb.service; enabled; vendor preset: enabled)
   Active: active (running) since Mon 2026-08-17 13:06:52 IST; 15min ago
     Docs: man:mariabdb(8)
           https://mariadb.com/kb/en/library/systemd/
  Main PID: 8306 (mariabdb)
    Status: "Taking your SQL requests now..."
     Tasks: 9 (limit: 22423)
    Memory: 68.9M
       CPU: 613ms
    CGroup: /system.slice/mariadb.service
            └─8306 /usr/sbin/mariabdb
```

```

sunith@sunith-virtual-machine:~$ mariadb --version
mariadb Ver 15.1 Distrib 10.6.23-MariaDB, for debian-linux-gnu (x86_64) using Editline wrapper
sunith@sunith-virtual-machine:~$ mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 42
Server version: 10.6.23-MariaDB-0ubuntu0.22.04.1 Ubuntu 22.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE companydb;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| companydb |
| information_schema |
| my_database |
| mysql |
| performance_schema |
| sys |
+-----+
6 rows in set (0.006 sec)

MariaDB [(none)]> CREATE USER 'sunith'@'localhost' IDENTIFIED BY 'Password@123';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON companydb.* TO 'sunith'@'localhost';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> SELECT User,Host FROM mysql.user;
+-----+-----+
| User | Host |
+-----+-----+
| mariadb.sys | localhost |
| my_user | localhost |
| mysql | localhost |
| root | localhost |
| sunith | localhost |
+-----+-----+
5 rows in set (0.007 sec)

MariaDB [(none)]> EXIT;
Bye

```

```

sunith@sunith-virtual-machine:~$ mysql -u sunith -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 48
Server version: 10.6.23-MariaDB-0ubuntu0.22.04.1 Ubuntu 22.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> USE companydb;
Database changed
MariaDB [companydb]> CREATE TABLE employees(
  ->     id INT AUTO_INCREMENT PRIMARY KEY,
  ->     name VARCHAR(100),
  ->     department VARCHAR(50),
  ->     salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.038 sec)

MariaDB [companydb]> SHOW TABLES;
+-----+
| Tables_in_companydb |
+-----+
| employees |
+-----+
1 row in set (0.001 sec)

```

```

MariaDB [companydb]> SELECT * FROM employees;
+-----+-----+-----+-----+
| id | name | department | salary |
+-----+-----+-----+-----+
| 1 | John | IT | 50000.00 |
| 2 | Sara | HR | 45000.00 |
+-----+-----+-----+-----+
2 rows in set (0.001 sec)

MariaDB [companydb]> exit
Bye
sunith@sunith-virtual-machine:~$ sudo apt install ufw -y
[sudo] password for sunith:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 447 not upgraded.
sunith@sunith-virtual-machine:~$ sudo ufw enable
Firewall is active and enabled on system startup
sunith@sunith-virtual-machine:~$ sudo ufw allow 22
Rule added
Rule added (v6)
sunith@sunith-virtual-machine:~$ 3306
3306: command not found
sunith@sunith-virtual-machine:~$ mysqldump -u root -p companydb > companydb_backup.sql
Enter password:
sunith@sunith-virtual-machine:~$ sudo systemctl enable mariadb
Synchronizing state of mariadb.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable mariadb
sunith@sunith-virtual-machine:~$
sunith@sunith-virtual-machine:~$ ls -lh
total 40K
-rw-rw-r-- 1 sunith sunith 2.2K Aug 17 13:41 companydb_backup.sql

```

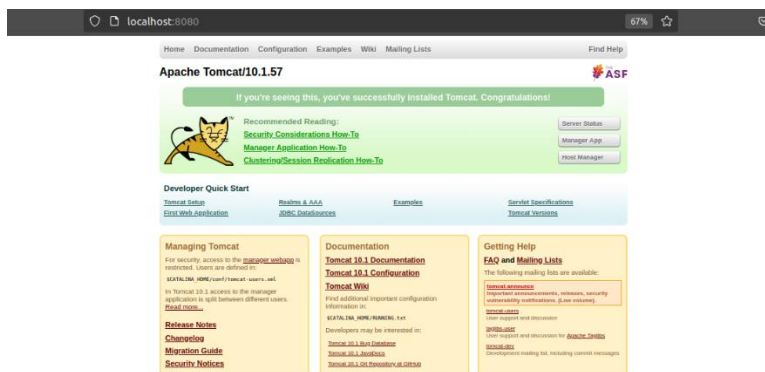
Conclusion: Successfully Implementation is Completed

6. Set up a Application Server (e.g.,Apache Tomcat)

Objective: To install and configure Apache Tomcat as an application server on Linux

Purpose: Apache Tomcat is commonly used to run Java-based web applications such as applications packaged as WAR files.

Implementation:



Conclusion: Successfully Implementation is Completed

7. create a service file for Apache Tomcat.(Should execute by using systemctl command)

Objective: To create a systemd service file for Apache Tomcat so that Tomcat can be controlled using systemctl.

Purpose: A systemd service file allows administrators to start, stop, restart, enable, and check the status of Tomcat using standard Linux service-management commands.

Implementation:

```
sunith@sunith-virtual-machine:/tmp$ cd /tmp
wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.57/bin/apache-tomcat-10.1.57.tar.gz
--2026-08-17 16:10:32-- https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.57/bin/apache-tomcat-10.1.57.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14400273 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-10.1.57.tar.gz'

apache-tomcat-10.1.57.tar.gz      100%[=====] 13.73M  30.8MB/s

2026-08-17 16:10:33 (30.8 MB/s) - 'apache-tomcat-10.1.57.tar.gz' saved [14400273/14400273]

sunith@sunith-virtual-machine:/tmp$ sudo mkdir -p /opt/tomcat
sudo tar -xzf apache-tomcat-10.1.57.tar.gz -C /opt/tomcat --strip-components=1
sunith@sunith-virtual-machine:/tmp$ ls /opt/tomcat
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
sunith@sunith-virtual-machine:/tmp$ cd /opt/tomcat/bin
sudo ./startup.sh
bash: cd: /opt/tomcat/bin: Permission denied
sudo: ./startup.sh: command not found
sunith@sunith-virtual-machine:/tmp$ sudo chown -R sunith:sunith /opt/tomcat
sunith@sunith-virtual-machine:/tmp$ ls -ld /opt/tomcat
drwxr-xr-x 9 sunith sunith 4096 Aug 17 16:10 /opt/tomcat
sunith@sunith-virtual-machine:/tmp$ cd /opt/tomcat/bin
./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
tomcat started.
sunith@sunith-virtual-machine:/opt/tomcat/bin$ ps aux | grep tomcat
sunith  49964  2.2  3.5 3374332 105940 pts/0  Sl   16:12   0:04 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener.UNMASK=0027 --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED -classpath /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat -Dcatalina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp org.apache.catalina.startup.Bootstrap start
sunith  50832  0.0  0.0  9080  2688 pts/0    S+   16:16   0:00 grep --color=auto tomcat
sunith@sunith-virtual-machine:/opt/tomcat/bin$ ss -ltnp | grep 8080
LISTEN 0      100        *:8080     :*      users:((("java",pid=49964,fd=43))
sunith@sunith-virtual-machine:/opt/tomcat/bin$ ls
bootstrap.jar catalina-tasks.xml commons-daemon.jar configtest.sh digest.sh migrate.bat setclasspath.sh startup.bat tomcat-native.tar.gz version.bat
atalina.bat  ciphers.bat  commons-daemon-native.tar.gz daemon.sh  makebase.bat  migrate.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
atalina.sh   ciphers.sh   configtest.bat digest.bat  makebase.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
sunith@sunith-virtual-machine:/opt/tomcat/bin$ pwd
/opt/tomcat/bin
sunith@sunith-virtual-machine:/opt/tomcat/bin$ cd ..
sunith@sunith-virtual-machine:/opt/tomcat$ cd ..
bash: cd: ..: No such file or directory
sunith@sunith-virtual-machine:/opt/tomcat$ cd ..
sunith@sunith-virtual-machine:/opt$ ls
tomcat
sunith@sunith-virtual-machine:/opt$ cd tomcat/
sunith@sunith-virtual-machine:/opt/tomcat$ ls
BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
```

```
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking

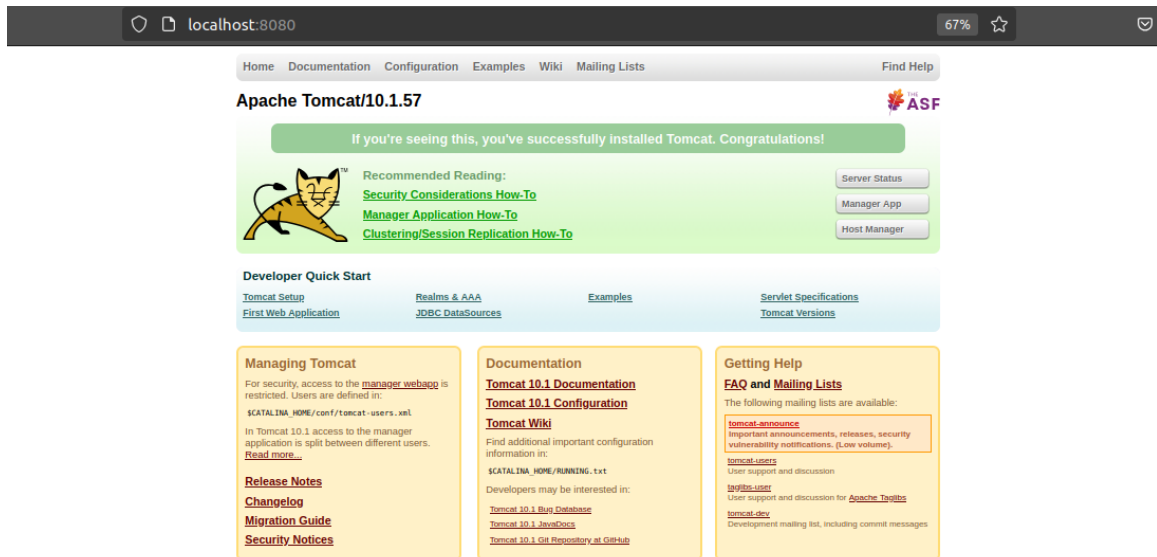
User=tomcat
Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-17-openjdk-and64"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure

[Install]
WantedBy=multi-user.target
```



Conclusion: Implementation Successfully completed

8. Print specific columns from a delimited file.

Objective: To extract and print specific columns from a file using the awk command.

Purpose: awk is useful for processing structured text files such as CSV, colon-separated files, and other delimited data.

Implementation:

```
sunith@sunith-virtual-machine:~$ ls
companydb_backup.sql  Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
sunith@sunith-virtual-machine:~$ vim employees.txt
sunith@sunith-virtual-machine:~$ cat employees.txt
101,Sunith,DevOps,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
103,Kiran,DevOps,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ awk -F',' '{print $2, $3}' employees.txt
Sunith DevOps
Ravi Linux
Kiran DevOps
Arun Testing
Anil Linux
Rahul Testing
sunith@sunith-virtual-machine:~$ awk -F',' '{print $2, $4}' employees.txt
Sunith 50000
Ravi 45000
Kiran 60000
Arun 40000
Anil 55000
Rahul 48000
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully implementation is completed

9. Filter and print lines based on a specific pattern or condition.

Objective: To search for and print only the lines that match a specified pattern or condition.

purpose: Filtering is useful when working with large files where only specific records or lines are required

Implementation:

```
sunith@sunith-virtual-machine:~$ awk -F',' ' $3=="DevOps" {print}' employees.txt
101,Sunith,DevOps,50000,Hyderabad
103,Kiran,DevOps,60000,Bangalore
sunith@sunith-virtual-machine:~$ awk -F',' ' $4 > 50000 {print}' employees.txt
103,Kiran,DevOps,60000,Bangalore
105,Anil,Linux,55000,Delhi
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully implementation is completed

10. Calculate and print the average, sum, or other statistics of a column.

Objective: To calculate numerical statistics such as sum and average from a particular column using awk.

Purpose: awk can process numerical data and perform calculations directly from text files.

Implementation:

```
sunith@sunith-virtual-machine:~$ awk -F',' '{sum += $4} END {print "Total Salary:", sum}' employees.txt
Total Salary: 298000
sunith@sunith-virtual-machine:~$ awk -F',' '{sum += $4; count++} END {print "Average Salary:", sum/count}' employees.txt
Average Salary: 49666.7
sunith@sunith-virtual-machine:~$ awk -F',' 'BEGIN {min=999999999} {if($4<min) min=$4} END {print "Minimum Salary:",min}' employees.txt
Minimum Salary: 40000
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully Implementation is completed

11. Perform string manipulation, such as extracting substrings or changing case.

Objective: To perform string operations such as extracting substrings, changing case, and modifying text using Linux commands and awk.

Purpose: String manipulation is useful when processing usernames, filenames, application data, log files, and other text-based information.

Implementation:

```
sunith@sunith-virtual-machine:~$ awk -F',' '{print toupper($2)}' employees.txt
SUNITH
RAVI
KIRAN
ARUN
ANIL
RAHUL
sunith@sunith-virtual-machine:~$ awk -F',' '{print tolower($2)}' employees.txt
sunith
ravi
kiran
arun
anil
rahul
sunith@sunith-virtual-machine:~$ awk -F',' '{print substr($2,1,3)}' employees.txt
Sun
Rav
Kir
Aru
Ani
Rah
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully Implementation Is Completed

12.Count the occurrences of a specific pattern in a file.

Objective: To count how many times a particular pattern occurs in a file.

Purpose: Pattern counting is useful for analyzing logs, monitoring errors, counting events, and checking application activity.

Implementation:

```
sunith@sunith-virtual-machine:~$ grep -c "DevOps" employees.txt
2
sunith@sunith-virtual-machine:~$ grep -c "Linux" employees.txt
2
sunith@sunith-virtual-machine:~$ awk -F',' '$3=="DevOps" {count++} END {print count}' employees.txt
2
sunith@sunith-virtual-machine:~$ grep -o "DevOps" employees.txt | wc -l
2
sunith@sunith-virtual-machine:~$
```

Conclusion: successfully Implementation is Completed

13. Sort lines based on a specific field or column.

Objective: To sort file contents according to a specific field or column

Purpose: Sorting makes structured data easier to analyze and organize.

Implementation:

```
sunith@sunith-virtual-machine:~$ sort -t',' -k4,4n employees.txt
104,Arun,Testing,40000,Hyderabad
102,Ravi,Linux,45000,Chennai
106,Rahul,Testing,48000,Chennai
101,Sunith,DevOps,50000,Hyderabad
105,Anil,Linux,55000,Delhi
103,Kiran,DevOps,60000,Bangalore
sunith@sunith-virtual-machine:~$ sort -t',' -k4,4nr employees.txt
103,Kiran,DevOps,60000,Bangalore
105,Anil,Linux,55000,Delhi
101,Sunith,DevOps,50000,Hyderabad
106,Rahul,Testing,48000,Chennai
102,Ravi,Linux,45000,Chennai
104,Arun,Testing,40000,Hyderabad
sunith@sunith-virtual-machine:~$ sort -t',' -k2,2 employees.txt
105,Anil,Linux,55000,Delhi
104,Arun,Testing,40000,Hyderabad
103,Kiran,DevOps,60000,Bangalore
106,Rahul,Testing,48000,Chennai
102,Ravi,Linux,45000,Chennai
101,Sunith,DevOps,50000,Hyderabad
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully implementation is completed

14. Merge multiple files based on a common field or column.

Objective: To combine information from multiple files using a common field or key.

Purpose: Merging files is useful when information is stored in separate datasets and needs to be combined for analysis or reporting.

Implementation:

```

sunith@sunith-virtual-machine:~$ sort -t',' -k1,1 employees.txt > employees_sorted.txt
sunith@sunith-virtual-machine:~$ sort -t',' -k1,1 departments.txt > departments_sorted.txt
sunith@sunith-virtual-machine:~$ join -t',' -1 1 -2 1 employees_sorted.txt departments_sorted.txt
101,Sunith,DevOps,50000,Hyderabad,DevOps Team
102,Ravi,Linux,45000,Chennai,Linux Team
103,Kiran,DevOps,60000,Bangalore,DevOps Team
104,Arun,Testing,40000,Hyderabad,Testing Team
105,Anil,Linux,55000,Delhi,Linux Team
106,Rahul,Testing,48000,Chennai,Testing Team
sunith@sunith-virtual-machine:~$ cat employees.txt
101,Sunith,DevOps,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
103,Kiran,DevOps,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ cat departments.txt
101,DevOps Team
102,Linux Team
103,DevOps Team
104,Testing Team
105,Linux Team
106,Testing Team
sunith@sunith-virtual-machine:~$ █

```

Conclusion: Successfully Implementation Is Completed

15. Substitute text in a file using search and replace.

Objective: To search for specific text and replace it with new text using the sed command.

Purpose: Search and replace is useful for modifying configuration files, application files, scripts, and large text files.

Implementation:

```

sunith@sunith-virtual-machine:~$ sed 's/DevOps/Cloud/g' employees.txt
101,Sunith,Cloud,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed -i 's/DevOps/Cloud/g' employees.txt
sunith@sunith-virtual-machine:~$ cp employees.txt employees_backup.txt
sunith@sunith-virtual-machine:~$ sed -i 's/DevOps/Cloud/g' employees.txt
sunith@sunith-virtual-machine:~$ sed -i 's/DevOps/Cloud/g' employees.txt
sunith@sunith-virtual-machine:~$ cat employees
employees_backup.txt employees_sorted.txt employees.txt
sunith@sunith-virtual-machine:~$ cat employees
cat: employees: No such file or directory
sunith@sunith-virtual-machine:~$ cat employees.txt
101,Sunith,Cloud,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ cat employees_sorted.txt
101,Sunith,DevOps,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
103,Kiran,DevOps,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ █

```

Conclusion: Successfully Implementation IS Completed

16. Delete specific lines based on a pattern or line number.

Objective: To delete specific lines from a file based on a line number or matching pattern.

Purpose: Deleting unwanted lines is useful when cleaning configuration files, logs, scripts, and datasets.

Implementation:

```
sunith@sunith-virtual-machine:~$ sed '3d' employees.txt
101,Sunith,Cloud,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed '3d' employees.txt
101,Sunith,Cloud,50000,Hyderabad
102,Ravi,Linux,45000,Chennai
104,Arun,Testing,40000,Hyderabad
105,Anil,Linux,55000,Delhi
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed '/Linux/d' employees.txt
101,Sunith,Cloud,50000,Hyderabad
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed -i '/Linux/d' employees.txt
sunith@sunith-virtual-machine:~$
```

Conclusion: Successfully Implementation is Completed

17. Append or insert text before or after a specific pattern or line.

Objective: To insert or append text at a specific location in a file using sed.

Purpose: This is useful for modifying configuration files and automatically adding required settings or entries.

Implementation:

```
sunith@sunith-virtual-machine:~$ sed '2i New Employee Record' employees.txt
101,Sunith,Cloud,50000,Hyderabad
New Employee Record
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
```

```

sunith@sunith-virtual-machine:~$ sed '2a New Employee Record' employees.txt
101,Sunith,Cloud,50000,Hyderabad
103,Kiran,Cloud,60000,Bangalore
New Employee Record
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed '/DevOps/i --- DevOps Employee ---' employees.txt
101,Sunith,Cloud,50000,Hyderabad
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ sed '/DevOps/a --- End DevOps Record ---' employees.txt
101,Sunith,Cloud,50000,Hyderabad
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$

```

Conclusion: Successfully Implementation Is Completed

18. Print only specific lines from a file.

Objective: To display only selected lines from a file using commands such as sed, head, tail, and awk.

Purpose: Printing specific lines is useful when examining large files without displaying the complete contents.

Implementation:

```

sunith@sunith-virtual-machine:~$ sed -n '3p' employees.txt
104,Arun,Testing,40000,Hyderabad
sunith@sunith-virtual-machine:~$ sed -n '2,4p' employees.txt
103,Kiran,Cloud,60000,Bangalore
104,Arun,Testing,40000,Hyderabad
106,Rahul,Testing,48000,Chennai
sunith@sunith-virtual-machine:~$ █

```

Conclusion: Successfully Implementation is Completed

19. Copy file from linux to windows machine

Objective: To transfer files from a Linux machine to a Windows machine using a secure file-transfer method.

Purpose: File transfer is commonly required when moving scripts, configuration files, logs, backups, or other data between Linux and Windows systems.

Implementation:

```
HP@Ashish MINGW64 ~/Downloads
$ scp -i Devops.pem /c/Users/HP/Downloads/introduction-devops.pptx ec2-user@44.223.93.116:/tmp
The authenticity of host '44.223.93.116 (44.223.93.116)' can't be established.
ED25519 key fingerprint is SHA256:hrTw/86pRpkCg8BoU+06GoS1dLVnd6dpjnoXbcVzwYg.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: ec2-44-223-93-116.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.223.93.116' (ED25519) to the list of known hosts.
introduction-devops.pptx 100% 17MB 3.3MB

[ec2-user@ip-172-31-12-111 ~]$ ls /tmp/
introduction.tar
introduction-devops.pptx
systemd-private-c3cc0bcfb87d4bfaa02ca743f38c8c05-chrond.service-heMYr7
systemd-private-c3cc0bcfb87d4bfaa02ca743f38c8c05-dbus-broker.service-XDs0vs
systemd-private-c3cc0bcfb87d4bfaa02ca743f38c8c05-policy-routes@ens5.service-ZcEyzj
systemd-private-c3cc0bcfb87d4bfaa02ca743f38c8c05-systemd-logind.service-IBXv3Z
systemd-private-c3cc0bcfb87d4bfaa02ca743f38c8c05-systemd-resolved.service-SFlydP
[ec2-user@ip-172-31-12-111 ~]$ |

PS C:\Users\HP> scp -i "C:\Users\HP\Downloads\Devops.pem" ec2-user@44.223.93.116:/home/ec2-user/introduction.tar "C:\Users\HP\Downloads\
introduction.tar"
PS C:\Users\HP> | 100% 17MB 588.5KB/s 00:29
```

Conclusion: Successfully Implementation is Completed

20.5 use cases for AWK and 5 use cases for sed

Objective: To understand and demonstrate common practical uses of the awk command for processing structured text and data.

Purpose: awk is particularly useful for column processing, filtering, calculations, and text manipulation

Implementation:

```
sunith@sunith-virtual-machine:~$ awk -F',' '{print $2,$4}' employees.txt
Sunith 50000
Kiran 60000
Arun 40000
Rahul 48000
sunith@sunith-virtual-machine:~$ awk -F',' '$3=="DevOps" {print}' employees.txt
sunith@sunith-virtual-machine:~$ awk -F',' '{sum += $4} END {print sum}' employees.txt
198000
sunith@sunith-virtual-machine:~$ awk -F',' '{print toupper($2)}' employees.txt
SUNITH
KIRAN
ARUN
RAHUL
sunith@sunith-virtual-machine:~$ |
```

Conclusion: Successfully Implementation is completed